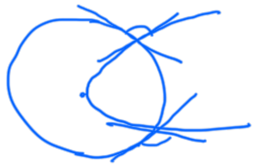
Orthogonal trajectory

Definition: The angle of intersection between two curves is defined as the angle between the tangents of the curves at the point of intersection.



Orthogonal trajectory: A family of curves that intersects a given family of curves at right angles is called orthogonal trajectory.

Method (1) Find the diff. eq. of a given family of curves (1-parameter)

(2) Replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$ and solve the diff. eq.

Ex. $x^2 + y^2 = a^2$, a – arbitrary constants

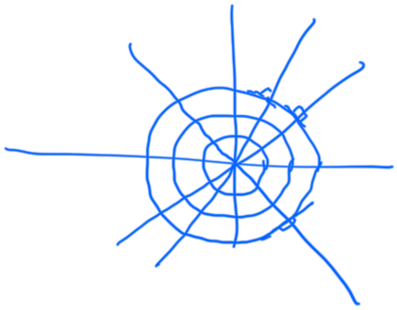
Soln: $2x + 2y\frac{dy}{dx} = 0 \Rightarrow x + y\frac{dy}{dx} = 0$.

Replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$,

$$x + y\left(-\frac{dx}{dy}\right) = 0 \Rightarrow \frac{dx}{x} = \frac{dy}{y}$$

On integrating, $\ln y = \ln x + \ln c$

$\Rightarrow \boxed{y = cx}$ → orthogonal trajectory.

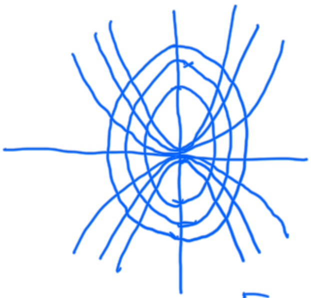


Ex $y^2 = ax$

Sol: $2y\dfrac{dy}{dx} = a \Rightarrow 2y\dfrac{dy}{dx} = \dfrac{y^2}{x}$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{y}{2x}$$

Replace $\dfrac{dy}{dx}$ by $-\dfrac{dx}{dy}$,



$$-\frac{dx}{dy} = \frac{y}{2x} \Rightarrow 2x\,dx + y\,dy = 0$$

$$\Rightarrow \boxed{x^2 + \frac{y^2}{2} = c.}$$

Ex. Find orthogonal trajectories of

(1) $x^2 - y^2 = c$ (2) $y = 4x + c$ (3) $y = \dfrac{c}{x}$

(4) $y = 2x^2 + c$ (5) $y = ce^{-3x}$ (6) $4x^2 + y^2 = c.$

In Polar co-ordinate: (1) Find the diff. eq

(2) Replace $\dfrac{1}{r}\dfrac{dr}{d\theta}$ by $-r\dfrac{d\theta}{dr}$ and solve.

Ex $r = 2a\cos\theta$, a – arbitrary constant.

Sol: $\dfrac{dr}{d\theta} = -2a\sin\theta \Rightarrow \dfrac{dr}{d\theta} = -\dfrac{r\sin\theta}{\cos\theta}$

$$\Rightarrow \frac{1}{r}\frac{dr}{d\theta} = -\tan\theta$$

Replace $\dfrac{1}{r}\dfrac{dr}{d\theta} = -r\dfrac{d\theta}{dr}$, this gives, $-r\dfrac{d\theta}{dr} = -\tan\theta.$

$$\Rightarrow \frac{d\theta}{\tan\theta} = \frac{dr}{r}$$

$$\Rightarrow \ln r = \ln\sin\theta + \ln c$$

$$\Rightarrow \boxed{r = c\sin\theta.}$$

Existence and Uniqueness theorem for IVP

Initial value problem (IVP) for the first order diff eq:

$$\frac{dy}{dx} = f(x,y), \qquad y(a) = b.$$

<u>Q.1</u> Does solution of IVP always exists?

<u>Ans</u>: No.

<u>Ex</u>. $\underbrace{|y'|} + \underbrace{|y|} = 0, \qquad y(0) = 1.$

$$\Downarrow$$

$$|y'| = 0 \;\&\; |y| = 0$$

$$\Downarrow \qquad\qquad \Downarrow$$

$$y' = 0 \qquad\qquad y = 0$$

$$\Downarrow$$

$$y = c \Longrightarrow y = 0.$$

$\therefore$ $y = 0$ is the only solution of $|y'| + |y| = 0$ but it does not satisfy the initial condition $y(0) = 1$. Therefore, the given IVP has no solution.

<u>Q 2</u> If solution exists, is it unique?

<u>Ans</u>: Not necessary.

<u>Ex</u> $\frac{dy}{dx} = y^{1/3}, \quad y(0) = 0.$

Clearly, $y = 0$ is a solution.

$$\frac{dy}{y^{1/3}} = dx$$

$$\Rightarrow \frac{3}{2} y^{2/3} = x + c$$

$$\Rightarrow y = \left[\frac{2(x+c)}{3}\right]^{3/2}.$$

$\because y(0) = 0, \qquad 0 = \left(\frac{2c}{3}\right)^{3/2} \Rightarrow c = 0$

$\therefore y = \left(\frac{2x}{3}\right)^{3/2}; \quad x \geq 0$ is another solution.

Ex. show that the above IVP has infinitely many solution.

Hint: Take $y = \begin{cases} 0 & 0 \leq x < a \\ \left(\frac{2}{3}(x-a)\right)^{3/2}, & x \geq a \end{cases}$, $\quad a > 0.$

Is y differentiable ?

Ex. Show that IVP $\frac{dy}{dx} = \frac{3x^2 + 4x + 2}{2y - 2}$; $y(0) = 1$.

has more than one solution.